

Notes: Dichev and Piotroski (JF, 2001)

The Long-Run Stock Returns Following Bond Ratings Changes

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1 One-sentence summary

Dichev and Piotroski (2001) use a comprehensive sample of Moody's bond rating changes from 1970 to 1997 and show that stock returns do not reliably drift after upgrades, but stocks earn large negative abnormal returns after downgrades - especially small, low-credit-quality firms - suggesting that the equity market underreacts to the negative information in debt-rating downgrades.

2 Big picture

The paper studies whether the stock market fully incorporates the information contained in bond rating changes. A bond rating change is a public, salient, and economically meaningful signal about a firm's credit quality. If equity prices are informationally efficient, the stock-price reaction should occur around the announcement, and there should be little abnormal return afterward.

The paper's main finding is asymmetric. Upgrades produce little reliable long-run abnormal performance. Downgrades, however, are followed by substantial negative abnormal stock returns. In the first year after a downgrade, abnormal returns are roughly -10% to -14% , depending on sample and methodology.

The authors interpret the pattern as evidence of delayed market reaction rather than compensation for systematic risk. The evidence is strongest for firms where information problems are likely to be severe: small firms and firms with noninvestment-grade debt.

3 Research question

The paper asks four related questions.

1. **Do stock prices fully react to bond rating changes?** The main empirical test is whether abnormal returns continue after the rating-change announcement.
2. **Are upgrades and downgrades symmetric?** The authors separately study upgrades and downgrades because good and bad credit news may have different information content and different investor responses.
3. **Where is the return drift strongest?** The paper studies heterogeneity by parent versus subsidiary debt, firm size, credit quality, and recent earnings news.
4. **What explains the drift after downgrades?** The paper evaluates whether the negative returns look like risk compensation, sample-specific bad luck, or market underreaction.

4 Incremental contribution

4.1 A much more comprehensive sample

Earlier studies of bond rating changes used relatively small hand-collected samples, often limited to NYSE and AMEX firms. This paper uses Moody's Default Risk Service and builds a sample that includes essentially all Moody's bond rating changes for U.S. firms from 1970 to 1997.

The starting universe contains 49,339 Moody’s bond rating changes. After collapsing simultaneous changes for the same firm and matching to CRSP and book-to-market data, the final sample contains 4,727 rating changes: 1,787 upgrades and 2,940 downgrades. The sample includes smaller, Nasdaq-listed, and lower-credit-quality firms that are more likely to display delayed price reaction.

4.2 A long-run event-study perspective

Prior work mainly focused on announcement-window or short-window price effects. This paper studies three-month, six-month, first-year, second-year, and third-year returns after the rating change. This allows the authors to ask whether the market’s reaction is complete or whether drift persists.

The paper reports both cumulative abnormal returns (CARs) and buy-and-hold abnormal returns (BHARs), controlling for size and book-to-market. This is important because long-run return measurement is sensitive to benchmark choice.

4.3 Parent versus subsidiary distinction

The paper distinguishes between parent-issued and subsidiary-issued debt. This matters because a subsidiary rating change may not map cleanly into the equity value of the traded parent company, especially when the subsidiary debt is explicitly or implicitly supported by the parent.

The return effects are cleaner and stronger for parent-issued debt. This distinction sharpens interpretation and is one reason the parent-firm results receive special attention.

4.4 Mechanism evidence beyond raw drift

The paper does not simply document negative returns after downgrades. It also shows that downgrades predict future deteriorations in fundamentals, including subsequent rating downgrades and lower future profitability. It then shows that stock prices react negatively around later earnings announcements, which suggests that investors are surprised by the predictable future deterioration.

This mechanism evidence supports an underreaction interpretation: the downgrade contains information about future firm performance, but the market does not fully incorporate it immediately.

5 Data and sample construction

5.1 Source and universe

The bond rating data come from Moody’s Default Risk Service, which contains Moody’s bond rating changes between 1970 and 1997. Because a single firm can have multiple debt issues changed on the same date, the authors collapse simultaneous issue-level changes into one firm-level event by retaining the most senior issue available.

The sample construction proceeds as follows:

- 49,339 total Moody’s debt rating changes.

- 7,838 unique firm-date events after eliminating redundant same-firm same-date issue changes.
- 5,493 events matched to CRSP stock returns.
- 4,727 events with both size and book-to-market data.

The final sample has 2,399 parent-issued debt events and 2,328 subsidiary-issued debt events. Among parent issues, there are 863 upgrades and 1,536 downgrades; among subsidiary issues, there are 924 upgrades and 1,404 downgrades.

5.2 Firm characteristics

Upgrade firms are larger and have lower book-to-market ratios than downgrade firms. Downgrades are more likely to involve distressed or marginal firms. Because size and book-to-market are known predictors of returns, the paper controls for both when calculating abnormal performance.

Most events involve NYSE-listed firms, but the sample also includes AMEX and Nasdaq firms. This breadth is important because delayed reaction is often strongest among smaller and less-followed firms.

5.3 Rating-change distribution

Downgrades outnumber upgrades throughout the sample. The upgrade-downgrade mix varies by macroeconomic conditions: downgrades are more frequent during weaker economic periods, while upgrades become relatively more common during expansions.

The transition matrix shows that most rating changes are either within the same broad rating class or across one class. Only a small fraction involve large jumps of two or more broad classes. About 40 percent of new ratings are below Baa, the traditional cutoff for investment-grade bonds.

6 Abnormal return methodology

6.1 Announcement returns

For comparison with prior work, the paper reports three-day announcement-period returns around the rating change. Downgrades have significantly negative announcement returns. Upgrades have small positive announcement returns.

The key research question is whether the market reaction continues beyond the announcement period.

6.2 CAR method

For each month, the authors form 25 value-weighted benchmark portfolios by sorting all eligible CRSP firms into five size groups and five book-to-market groups. The size breakpoints are based on NYSE firms. A firm's monthly abnormal return is its return minus the return on the matched size-book-to-market benchmark portfolio.

The cumulative abnormal return over horizon H is:

$$CAR_{i,H} = \sum_{m=1}^H (R_{i,m} - R_{i,m}^{bench}).$$

6.3 BHAR method

The buy-and-hold abnormal return compounds the firm's return and subtracts the compounded benchmark return:

$$BHAR_{i,H} = \prod_{m=1}^H (1 + R_{i,m}) - \prod_{m=1}^H (1 + R_{i,m}^{bench}).$$

The paper reports both CARs and BHARs because each has different strengths. CARs have cleaner statistical properties, while BHARs match the investor experience of buying and holding after the event.

6.4 Fama-MacBeth regressions

The paper also runs monthly cross-sectional regressions using all CRSP firms with size and book-to-market data:

$$RETURN_i = a + \beta_1 \log(MVE_i) + \beta_2 \log(BM_i) + \beta_3 Event_i + \varepsilon_i.$$

The event indicator equals one if the firm was upgraded or downgraded within a specified prior window. This approach helps address cross-sectional dependence in event returns. The paper reports both equal-weighted and precision-weighted averages of the monthly coefficients.

7 Main results

7.1 Upgrades: little reliable drift

For upgrades, the paper finds no robust evidence of abnormal long-run returns in the portfolio tests. Some Fama-MacBeth specifications show modest positive abnormal returns, but the authors treat this evidence cautiously because it is weaker and less consistent than the downgrade evidence.

The economic message is that good credit news appears to be incorporated relatively quickly or is less informative for equity investors.

7.2 Downgrades: large negative abnormal returns

Downgrades are followed by substantial negative abnormal returns. For the full sample, one-year abnormal returns are approximately -10% . For parent-issued debt, the corresponding first-year abnormal returns are roughly -13% to -14% . The returns are statistically significant and economically large.

The drift is strongest in the first months and quarters after the announcement and weakens in the second and third years. This timing is important because it looks more like delayed reaction to information than a permanent risk premium.

7.3 Parent-issued debt gives stronger effects

The negative drift after downgrades is larger for parent-issued debt than for the all-firms sample. This is intuitive because a parent debt downgrade is more directly linked to the publicly traded equity claim being studied.

Subsidiary debt changes may contain noisier information about parent equity value, especially if the parent supports subsidiary debt or if subsidiary performance is only part of the parent's consolidated value.

8 Heterogeneity and mechanism

8.1 Small and low-credit-quality firms

The downgrade drift is especially pronounced for small firms with noninvestment-grade debt. In Table VI, the one-year CAR for small noninvestment-grade parent downgrades is about -21% , with a very large t-statistic. This is the strongest subgroup result in the paper.

This pattern is central to the interpretation. Small, distressed, low-credit-quality firms are exactly the firms for which investor attention, analyst following, and information processing are likely to be weakest. The drift therefore looks like underreaction rather than a broad systematic risk premium.

8.2 Not explained by post-earnings-announcement drift

Because profitability helps determine bond ratings, downgrades may be correlated with poor earnings news. The paper therefore conditions on the sign of the most recent quarterly earnings change.

Downgrades underperform even after positive earnings changes. This means the downgrade drift is not simply a repackaged post-earnings-announcement drift.

8.3 Downgrades predict future deterioration

Downgrades predict future fundamentals. About one-quarter of downgraded parent firms receive another downgrade within 12 months. Downgraded firms also experience deteriorating future profitability.

This is important because it verifies that downgrades have forward-looking information content rather than simply summarizing past bad news.

8.4 Market surprise at later earnings announcements

If investors fully understood the future implications of downgrades, later earnings announcements should not systematically surprise them. The paper finds the opposite: downgraded firms have

negative returns around subsequent earnings announcements, especially small noninvestment-grade downgrades.

This result is powerful because short-window earnings announcement returns are less likely to be explained by omitted long-run risk factors. They point to delayed recognition of deteriorating fundamentals.

8.5 Persistent across years

The paper shows that downgrade underperformance occurs in nearly all years of the sample. CARs are negative in 26 of 28 years, and BHARs are negative in 27 of 28 years. The effect is also negative in each of three broad subperiods.

This persistence makes a systematic-risk explanation less compelling. A risk-based hedge should sometimes pay off, but the downgrade portfolio rarely earns positive abnormal returns.

9 Identification and interpretation

9.1 What the paper can show

The paper shows a robust conditional association: following a downgrade, especially for parent firms, small firms, and noninvestment-grade firms, subsequent stock returns are abnormally low after standard size and book-to-market adjustments.

The mechanism tests show that downgrades predict future deteriorations and that markets react negatively when those deteriorations are later revealed. This strengthens the case that the drift reflects underreaction.

9.2 What remains difficult

As with all long-run return studies, it is difficult to rule out every possible omitted risk factor. The paper is careful about this. It argues that the total pattern - limited duration, strongest effects among less-followed firms, negative earnings-announcement surprises, and persistence across almost all years - is hard to reconcile with a standard systematic-risk explanation.

9.3 Economic interpretation

A bond downgrade is not just a debt-market event. It is a public signal that a firm's future prospects and default risk have deteriorated. Equity investors react negatively at the announcement, but not enough. The subsequent negative drift reflects gradual incorporation of the bad news into stock prices.

10 How to read the paper's contribution

This paper is best read as evidence about market efficiency and investor underreaction. The core result is not merely that downgrade firms perform poorly. It is that they perform poorly *after* the downgrade, after controlling for size and book-to-market, and that later earnings-announcement reactions show the market was still surprised by information that downgrades helped predict.

The paper also contributes to the literature on asymmetric market reactions. Bad news events appear to produce stronger and more persistent abnormal returns than comparable good news events. This asymmetry is consistent with a broader pattern in finance and accounting: negative news often generates larger delayed price correction than positive news.

11 Tables and figures

11.1 Figure 1: Derivation of the sample of Moody's bond rating changes for 1970 to 1997

11.2 Table I: Descriptive Statistics for the Sample of Moody's Bond Ratings Changes

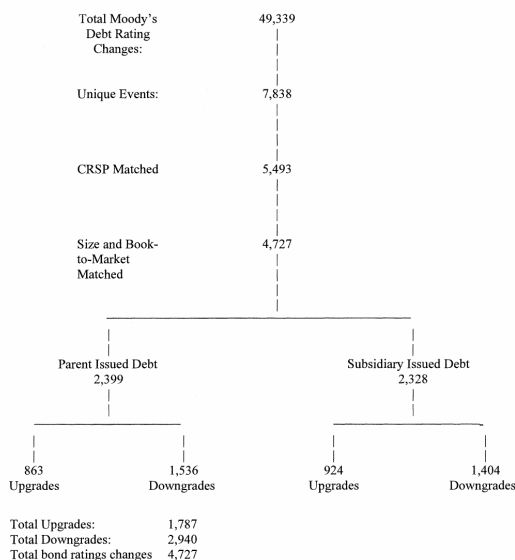


Figure 1. Derivation of the sample of Moody's bond rating changes for 1970 to 1997.

Figure 1: Derivation of the sample of Moody's bond rating changes for 1970 to 1997

Table I
Descriptive Statistics for the Sample of Moody's Bond Ratings Changes

Table I presents descriptive statistics for the full sample of bond ratings changes, spanning 1970 to 1997. Panel A presents the size and book-to-market characteristics. Market value of equity (MVE) equals total number of shares outstanding times beginning of the fiscal year stock price per share, and is presented in millions. Book-to-market (BM) equals the book value of common equity at the beginning of the fiscal year containing the bond rating change, scaled by MVE. Book values are from COMPUSTAT and Moody's Manuals. Firms with negative book value are eliminated from the descriptive statistics. Panel B presents a breakdown of the sample by exchange listing.

Panel A: Size and Book-to-Market Characteristics for All CRSP Firms, Upgrade Firms, and Downgrade Firms							
	Mean	Standard Deviation	5%	25%	50%	75%	95%
All CRSP Firms							
MVE	7,743	11,572	3	14	59	332	6,040
BM	3.89	61.45	0.005	0.34	0.71	1.33	3.46
Upgrade Firms							
MVE	4,065	8,347	75	424	1,343	3,614	16,400
BM	0.69	0.47	0.18	0.39	0.63	0.90	1.39
Downgrade Firms							
MVE	2,598	5,545	19	141	631	2,403	12,353
BM	1.12	1.30	0.28	0.59	0.89	1.30	2.64
Panel B: Exchange Listing							
	Upgrade Firms	Downgrade Firms	Total	Percent of Total			
NYSE	1,517	2,389	3,906	82.6%			
AMEX	60	198	258	5.5%			
NASDAQ	210	353	563	11.9%			
Total	1,787	2,940	4,727	100%			

Table I: Descriptive Statistics for the Sample of Moody's Bond Ratings Changes

Read: Figure 1 shows the sample funnel from 49,339 Moody’s debt rating changes to 4,727 usable observations with CRSP returns and size/book-to-market data. It also separates parent-issued and subsidiary-issued debt.

Economic message: The final sample is much broader than earlier hand-collected samples and contains enough downgraded firms to study long-run drift with meaningful power.

Read: Table I shows that upgrade firms are larger and have lower book-to-market than downgrade firms. Most observations are NYSE listed, but AMEX and Nasdaq firms are included.

Economic message: Size and book-to-market differences are large enough that benchmark adjustment is essential before interpreting post-rating-change returns.

11.3 Table II: The Distribution of Moody’s Bond Ratings Changes Over Time (Full Sample)

11.4 Table III: Bond Rating Change Matrix (Full Sample)

Table II
The Distribution of Moody’s Bond Ratings Changes Over Time (Full Sample)

Year	Number of Upgrades	Number of Downgrades	Number of Bond Ratings Changes	Percentage of Total Bond Ratings Changes
1970	9	12	21	0.44
1971	11	18	29	0.61
1972	16	7	23	0.49
1973	13	13	26	0.55
1974	22	40	62	1.31
1975	32	26	58	1.23
1976	18	14	32	0.68
1977	23	27	50	1.06
1978	19	10	29	0.61
1979	20	34	54	1.14
1980	14	46	60	1.27
1981	21	53	74	1.57
1982	36	144	180	3.81
1983	75	99	174	3.68
1984	105	132	237	5.01
1985	100	153	253	5.35
1986	93	247	340	7.19
1987	84	185	269	5.69
1988	91	218	309	6.54
1989	90	216	306	6.47
1990	65	322	387	8.19
1991	73	236	309	6.54
1992	94	164	258	5.46
1993	128	106	234	4.95
1994	113	100	213	4.51
1995	131	113	244	5.16
1996	173	95	268	5.67
1997	118	110	228	4.82
Total	1,787	2,940	4,727	100%

Table II: The Distribution of Moody’s Bond Ratings Changes Over Time (Full Sample)

Read: Table II shows that rating-change activity rises sharply after the early 1980s and varies strongly over time. Downgrades are especially common in recessionary periods such as 1990-1991.

Economic message: The sample spans different credit-cycle environments, which helps later tests ask whether downgrade underperformance is persistent or concentrated in only one subperiod.

Table III
Bond Rating Change Matrix (Full Sample)

This table presents descriptive evidence on the magnitude of Moody’s bond rating changes for the full sample of 4,727 observations between 1970 and 1997. Rows represent the original rating assigned by Moody’s, columns represent the new rating assigned by Moody’s after the change, and the number in each cell represents the number of observations that have the respective old and new bond rating. The diagonal of the matrix captures within-class rating changes. Moody’s did not have the finer within-class rating designations (1, 2, and 3) before April 20, 1982, and accordingly, there are no within-class changes before that date. The proportion of downgrades to total within-class rating changes is presented in the panel below the main table.

	New Bond Rating								
	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C
Old Bond Rating									
Aaa	0	71	4	0	0	0	0	0	0
Aa	35	183	312	6	1	0	0	0	0
A	1	149	629	433	27	2	0	1	0
Baa	1	6	264	544	234	14	0	0	0
Ba	0	0	5	189	262	255	15	1	0
B	1	0	2	16	168	428	269	27	1
Caa	0	1	2	5	4	51	4	88	2
Ca	0	0	0	3	0	1	3	0	6
C	0	0	0	0	0	1	0	0	0
Proportion of “within-class” rating changes which are downgrades (%)									
	n/a	68.3	52.8	53.3	50.8	62.9	75.0	n/a	n/a

Table III: Bond Rating Change Matrix (Full Sample)

Read: Table III shows that most rating changes are within the same broad rating class or across one adjacent class. About 40 percent of new ratings are below Baa.

Economic message: The sample has meaningful variation in credit quality, allowing the authors to test whether the drift is stronger where rating news is more severe and information problems are larger.

11.5 Table IV: The Long-Run Abnormal Stock Returns Following Moody's Bond Rating Changes

11.6 Table V: Fama-MacBeth Regressions of Monthly Returns on Size, Book-to-Market, and Bond Ratings Changes

Table IV
The Long-Run Abnormal Stock Returns Following Moody's Bond Rating Changes

Firm-specific BHARs are measured as the buy-and-hold raw return for the appropriate horizon minus the buy-and-hold return for a benchmark portfolio matched on size and book-to-market. Firm-specific CARs are the cumulations of monthly firm-specific returns minus the corresponding monthly return for the matching size and book-to-market portfolio. The abnormal returns presented in the table are the means of firm-specific abnormal returns for appropriate horizons. For comparative purposes, we also present the related three-day announcement period returns for these firms. *t*-statistics are presented in parentheses. Panel A presents results for all firms between 1970 and 1997 ($N = 4,727$); Panel B presents results for the parent-firm sample ($N = 2,399$).

	3-day Annc. Returns	3-month Returns	6-month Returns	1st-year Returns	2nd-year Returns	3rd-year Returns
Panel A: All Firms						
Upgrade firms						
BHAR	0.0048	0.0013	0.0001	0.0033	0.0101	-0.0237
(<i>t</i> -statistic)	(5.829)	(0.349)	(0.012)	(0.396)	(1.288)	(-3.043)
CAR	—	0.0019	0.0043	-0.0093	-0.0053	-0.0243
(<i>t</i> -statistic)	—	(0.546)	(0.879)	(-1.288)	(-0.719)	(-2.960)
Downgrade firms						
BHAR	-0.0197	-0.0395	-0.0662	-0.1014	-0.0368	-0.0423
(<i>t</i> -statistic)	(-11.47)	(-11.04)	(-7.939)	(-10.47)	(-3.842)	(-5.074)
CAR	—	-0.0402	-0.0647	-0.0957	-0.0448	-0.0339
(<i>t</i> -statistic)	—	(-8.108)	(-9.350)	(-10.251)	(-4.606)	(-3.744)
Panel B: Parent Firms						
Upgrade firms						
BHAR	0.0047	-0.0013	-0.0027	-0.0010	-0.0092	-0.0289
(<i>t</i> -statistic)	(3.643)	(-0.233)	(-0.333)	(-0.072)	(-0.762)	(-2.572)
CAR	—	0.0012	0.0024	-0.0120	-0.0272	-0.0306
(<i>t</i> -statistic)	—	(0.229)	(0.308)	(-1.097)	(-2.363)	(-2.500)
Downgrade firms						
BHAR	-0.0251	-0.0516	-0.0922	-0.1416	-0.0645	-0.0692
(<i>t</i> -statistic)	(-8.905)	(-6.924)	(-9.358)	(-9.728)	(-4.407)	(-5.521)
CAR	—	-0.0506	-0.0877	-0.1285	-0.0692	-0.0614
(<i>t</i> -statistic)	—	(-6.744)	(-8.010)	(-8.617)	(-4.606)	(-4.377)

Table IV: The Long-Run Abnormal Stock Returns Following Moody's Bond Rating Changes

Read: Table IV is the core return table. Upgrades show no reliable drift. Downgrades show large negative abnormal returns, especially for parent-issued debt, reaching about -13% to -14% in the first year.

Economic message: The market reaction to downgrades appears incomplete. The drift is large enough to be economically important, not just statistically detectable.

Table V
Fama-MacBeth Regressions of Monthly Returns on Size, Book-to-Market, and Bond Ratings Changes

This table presents the average coefficients and *t*-statistics (in parentheses) on UPGRADE and DOWNGRADE from the monthly estimation (January 1970 to December 1997) of the following cross-sectional regression:

$$\text{RETURN}_i = \alpha + \beta_1 \log(\text{MVE}_i) + \beta_2 \log(\text{BM}_i) + \beta_3 (\text{UPGRADE}_i \text{ or } \text{DOWNGRADE}_i)$$

RETURN is raw monthly return. MVE equals total number of shares outstanding times stock price per share. Book-to-market (BM) equals the book value of common equity, scaled by MVE. The indicator variable UPGRADE (DOWNGRADE) equals one if the firm's debt was upgraded (downgraded) within the specified time horizon preceding the monthly return, zero otherwise. Equal-weighted coefficients are the average coefficients from the 336 monthly regressions, and their *t*-statistics are equal to the average coefficient divided by its time-series standard error. Precision-weighted coefficients are defined identically, only first the monthly cross-sectional estimates are weighted by the inverse of their cross-sectional standard error.

	3-month	6-month	1st-year	2nd-year	3rd-year
Panel A: All Firms					
Upgrade					
Equal-weighted	0.0089 (3.972)	0.0049 (2.640)	0.0028 (2.193)	0.0020 (1.517)	0.0041 (2.756)
Precision-weighted	0.0080 (5.851)	0.0054 (5.019)	0.0036 (4.355)	0.0025 (2.996)	0.0039 (3.992)
Downgrade					
Equal-weighted	-0.0127 (-4.539)	-0.0098 (-4.257)	-0.0042 (-2.260)	-0.0024 (-1.285)	-0.0022 (-1.680)
Precision-weighted	-0.0157 (-6.801)	-0.0106 (-6.128)	-0.0059 (-4.387)	-0.0021 (-1.611)	-0.0026 (-2.215)
Panel B: Parent Firms					
Upgrade					
Equal-weighted	0.0075 (3.179)	0.0032 (1.435)	0.0024 (1.475)	-0.0001 (-0.088)	0.0029 (1.762)
Precision-weighted	0.0076 (4.781)	0.0046 (3.482)	0.0032 (3.359)	0.0008 (0.871)	0.0007 (0.351)
Downgrade					
Equal-weighted	-0.0195 (-5.297)	-0.0130 (-4.107)	-0.0080 (-3.283)	-0.0011 (-1.135)	-0.0017 (-1.351)
Precision-weighted	-0.0233 (-7.806)	-0.0148 (-6.826)	-0.0088 (-5.320)	-0.0016 (-1.429)	-0.0022 (-1.886)

Table V: Fama-MacBeth Regressions of Monthly Returns on Size, Book-to-Market, and Bond Ratings Changes

Read: Table V confirms the downgrade drift using Fama-MacBeth regressions. The downgrade coefficients are negative and significant, especially within the first year.

Economic message: The result is not an artifact of simple portfolio *t*-statistics or cross-sectional dependence in overlapping event returns.

11.7 Table VI: One-year Mean CARs Following Bond Rating Changes, Conditional on Firm Size and Credit Quality (Parent-issued Debt Only)

11.8 Table VII: One-Year Mean CARs Following Bond Rating Changes After Controlling for the Sign of the Post-Earnings-Announcement Drift (Parent-Issued Debt Only)

Table VI
One-year Mean CARs Following Bond Rating Changes, Conditional on Firm Size and Credit Quality (Parent-issued Debt Only)

This table presents one-year mean CARs following upgrades and downgrades, conditional on firm size (market value of equity) and credit quality (investment- or non-investment-grade). Firms are classified as small (large) if their market value of equity is less than (greater than or equal to) the median market value of equity for parent firms with a Moody's debt rating change in the same year. Bonds are classified as investment grade (noninvestment grade) if the new Moody's rating is Baa or better (Ba or lower). CARs are the cumulations of monthly firm-specific returns minus the corresponding monthly return for the matching size and book-to-market quintile from CRSP. *t*-statistics are in parentheses. *N* is the number of available observations for 1970 to 1997.

Firms	Investment Grade	Noninvestment Grade
Panel A: Upgrade Firms		
Large	CAR = -0.0355 <i>t</i> -stat. = (-2.713) <i>N</i> = 419	0.0840 (2.481) 99
Small	-0.0078 (-0.335) 124	-0.0129 (-0.458) 221
Panel B: Downgrade Firms		
Large	CAR = -0.0454 <i>t</i> -stat. = (-3.347) <i>N</i> = 525	-0.0601 (-1.323) 148
Small	-0.0991 (-3.618) 141	-0.2086 (-7.482) 722

Table VI: One-year Mean CARs Following Bond Rating Changes, Conditional on Firm Size and Credit Quality (Parent-issued Debt Only)

Read: Table VI shows that downgrade underperformance is concentrated in small, noninvestment-grade firms, with a one-year CAR around -21%. Upgrade results are weaker and less stable.

Economic message: The strongest drift appears exactly where limited attention and information-processing frictions are most plausible.

Table VII
One-Year Mean CARs Following Bond Rating Changes After Controlling for the Sign of the Post-Earnings-Announcement Drift (Parent-Issued Debt Only)

Table VII is organized as a 2×2 matrix that presents one-year mean CARs following upgrades and downgrades, conditional on the sign of the most recent preceding quarterly earnings change. CARs are the cumulations of monthly firm-specific returns minus the corresponding monthly return for the matched size and book-to-market quintiles from CRSP. *t*-statistics are in parentheses. *N* is the number of available observations for 1970 to 1997. Quarterly earnings changes are defined as income before extraordinary items less income from the same quarter in the previous year. The quarterly earnings realizations and announcement dates are obtained from COMPUSTAT's Quarterly Industrial tape, Compact Disclosure, annual reports, and the *Wall Street Journal Index*. The columns and the rows of the matrix are summed up to provide a reference for the return effect of bond rating changes and earnings changes, unconditional on each other. The entries in the cells of the matrix provide evidence about the abnormal return effects of bond ratings changes and earnings changes, conditional on each other.

	Upgrade Firms	Downgrade Firms	Unconditional Earnings Surprise Effect
Positive earnings surprise	CAR = -0.0032 <i>t</i> -stat. = (-0.233) <i>N</i> = 537	-0.1038 (-5.048) 599	-0.0563 (-4.412) 1136
Negative earnings surprise	-0.0298 (-1.659) 322	-0.1412 (-6.720) 910	-0.1121 (-6.889) 1232
Unconditional bond ratings change effect	-0.0132 (-1.202) 859	-0.1264 (-8.382) 1509	

Table VII: One-Year Mean CARs Following Bond Rating Changes After Controlling for the Sign of the Post-Earnings-Announcement Drift (Parent-Issued Debt Only)

Read: Table VII controls for the sign of the preceding quarterly earnings change. Downgraded firms underperform even when the most recent earnings change is positive.

Economic message: The downgrade drift is not simply the post-earnings-announcement drift in disguise; downgrades carry incremental negative information.

11.9 Table VIII: Subsequent Changes in Firm Fundamentals Conditional on Current Bond Rating Changes

11.10 Table IX: The Magnitude of Earnings Announcement Returns Following Bond Upgrades and Downgrades

Table VIII
Subsequent Changes in Firm Fundamentals Conditional on Current Bond Rating Changes

Table VIII presents subsequent bond rating actions and subsequent changes in return on equity (ROE), conditional on a previously observed bond rating action for parent firms for 1970 to 1997. For this table, a subsequent bond rating change is defined as a bond rating change within 12 months of the original bond rating change. ROE is measured as income before extraordinary items for the particular year (COMPUSTAT item 18) scaled by the beginning of the year book value of stockholder's equity (COMPUSTAT item 60), while Δ ROE is current year's ROE less the prior year's ROE. Firms with negative book values are eliminated from the ROE calculations.

Panel A: Subsequent Bond Ratings Actions Conditional on Original Bond Rating Action							
	Subsequent Bond Rating Action			Total			
	No Change	Upgrade	Downgrade				
Original bond rating action							
Upgrade %	89.5%	8.3%	2.2%	100%			
Number of obs.	772	72	19	863			
Downgrade %	73.5%	1.7%	24.8%	100%			
Number of obs.	1,129	26	381	1,536			
Panel B: Subsequent Changes in Profitability Conditional on Original Bond Rating Action							
	Mean (t-statistic)	Standard Deviation	5%	25%	50% (p-value)	75%	95%
Upgrade firms	0.022 (1.670)	0.357	-0.139	-0.003	0.003 (0.013)	0.033	0.214
Downgrade firms	-0.895 (-2.521)	11.96	-1.438	-0.175	-0.030 (0.000)	0.022	0.338

Table VIII: Subsequent Changes in Firm Fundamentals Conditional on Current Bond Rating Changes

Read: Table VIII shows that downgrades predict future bad news. Downgraded firms are much more likely to be downgraded again within 12 months and experience declines in profitability.

Economic message: Downgrades are forward-looking information events, not merely mechanical labels for past performance.

Table IX
The Magnitude of Earnings Announcement Returns Following Bond Upgrades and Downgrades

This table presents mean market-adjusted stock returns of upgraded and downgraded firms over the subsequent four quarterly earnings announcement periods for parent firms during the period of 1970 to 1997. Quarterly earnings announcement dates are from the COMPUSTAT Quarterly Industrial tape, the *Wall Street Journal* and the *Dow Jones News Retrieval Service*. Announcement returns are measured as the buy-and-hold returns earned over the five-day window $[-3, +1]$ surrounding each earnings announcement (date 0) less the value-weighted market index return over the same time interval. Mean returns for a particular quarter represents the average announcement return for those firms with returns available for that quarter. The total earnings announcement return for each firm (i.e., All Quarters) equals the sum of the individual quarterly earnings announcement returns. If announcement returns are not available for all four quarters, the total announcement return equals the sum of announcement returns over the available dates. The mean "All Quarters" return for each portfolio is the average of these firm-specific total earnings announcement returns. *t*-statistics are in parentheses. *N* is the number of available observation at each quarterly announcement. Panel A presents the announcement returns for the parent-issues sample. Panel B presents the announcement returns for the small, non-investment-grade parent issues. Panel C presents additional data on the characteristics of small non-investment-grade downgrades which miss earnings announcements after the first quarter. Performance delistings in Panel C are defined as in Shumway (1997), comprising CRSP delisting codes 500, 520, 551-574, 580, 584.

	First Quarter	Second Quarter	Third Quarter	Fourth Quarter	All Quarters
Mean Five-Day $[-3, +1]$ Returns					
Panel A: Earnings Announcement Returns for Parent Firms					
Upgrade firms	0.0028	0.0031	0.0030	-0.0036	0.0051
(<i>t</i> -statistic)	(1.508)	(1.590)	(1.490)	(-1.935)	(1.274)
<i>N</i>	843	822	822	822	843
Downgrade firms	-0.0101	-0.0033	-0.0034	0.0021	-0.0145
(<i>t</i> -statistic)	(-2.774)	(-1.155)	(-1.206)	(0.832)	(-2.779)
<i>N</i>	1449	1379	1342	1321	1449
Panel B: Earnings Announcement Returns for Small, Non-investment-grade Parent Issues					
Upgrade firms	0.0039	0.0006	0.0021	-0.0033	0.0033
(<i>t</i> -statistic)	(1.102)	(0.093)	(1.097)	(0.090)	(0.312)
<i>N</i>	221	211	210	210	221
Downgrade firms	-0.0183	-0.0128	-0.0058	0.0042	-0.0315
(<i>t</i> -statistic)	(-2.968)	(-2.242)	(-0.482)	(1.431)	(-3.178)
<i>N</i>	669	608	587	561	669

(continued)

Table IX: The Magnitude of Earnings Announcement Returns Following Bond Upgrades and Downgrades

Read: Table IX shows negative stock reactions around later earnings announcements after downgrades, especially for small noninvestment-grade firms.

Economic message: Investors appear surprised when future earnings deterioration arrives, supporting the underreaction interpretation.

11.11 Table IX–Continued: The Characteristics of Small, Non-investment-grade Parent Issues Missing At Least One Subsequent Earnings Announcement

11.12 Figure 2: The distribution of abnormal returns following parent downgrades over time

Table IX—Continued

Panel C: The Characteristics of Small, Non-investment-grade Parent Issues Missing At Least One Subsequent Earnings Announcement

C1: Announcement Returns						
	First Quarter	Second Quarter	Third Quarter	Fourth Quarter	All Quarters	One-year CAR
Returns	−0.0257	−0.0533	−0.0477	—	−0.0751	−0.4183
N	108	66	38	0	108	108

C2: Bankruptcies and Exchange Delistings						
	Bankrupt			Not bankrupt		
	49			59		
	Total number of observations missing at least one subsequent earnings announcement			108		
	Performance Delisting			No performance delisting		
	65			43		
	Available Delisting Return		No Delisting Return			
	13		52			

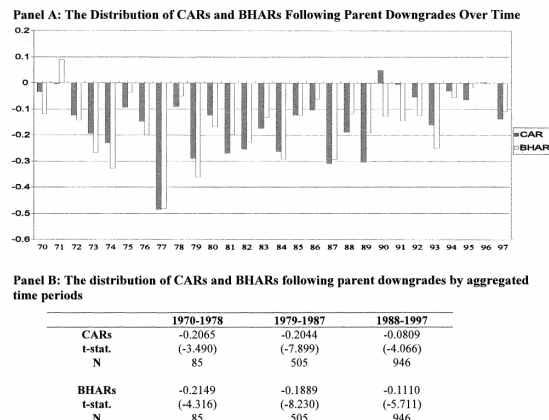


Figure 2. The distribution of abnormal returns following parent downgrades over time. Panel A presents the yearly distribution of CARs and BHARs following downgrades for parent firms from 1970 to 1997. Panel B presents the abnormal returns following downgrades aggregated in three time periods: 1970 to 1978, 1979 to 1987, and 1988 to 1997.

Table IX–Continued: Characteristics of small, noninvestment-grade parent issues missing at least one subsequent earnings announcement

Figure 2: The distribution of abnormal returns following parent downgrades over time

Read: The continuation of Table IX shows that many firms missing later earnings announcements are extremely distressed, frequently bankrupt or delisted for performance reasons.

Economic message: Measured earnings-announcement drift may understate the true bad-news effect because the worst downgraded firms often disappear from the observable earnings-announcement sample.

Read: Figure 2 shows downgrade abnormal returns by year and by broad subperiod. Abnormal returns are negative in nearly every year and remain negative across all major subperiods.

Economic message: The downgrade effect is persistent over time, which weakens the case that it is only a temporary sample-period artifact or a conventional risk premium.

12 Conclusion

12.1 Core conclusion

The paper's core conclusion is that equity markets do not fully and immediately incorporate the negative information in bond rating downgrades. Downgrades are followed by large and significant negative abnormal stock returns, while upgrades generate little reliable long-run drift.

12.2 Economic intuition

A downgrade is a credible signal from a specialized information intermediary that the firm's credit condition has worsened. Equity investors recognize some of this information at the announcement, but they underreact to its implications for future profitability and further credit deterioration. The correction then unfolds gradually through later returns and earnings announcements.

12.3 Incremental contribution revisited

The paper improves on prior work by using a much larger sample, including smaller and lower-credit-quality firms, studying horizons up to three years, distinguishing parent from subsidiary debt, and providing mechanism evidence using future rating changes, future profitability, and later earnings-announcement returns.

12.4 Implications

For market efficiency, the paper provides evidence of delayed reaction to publicly available credit information. For credit-market research, it shows that rating changes contain information relevant not only for bond investors but also for equity investors. For anomaly research, it reinforces the idea that bad-news events often generate stronger and more persistent drift than good-news events.

12.5 Limitations

The paper cannot completely rule out every possible omitted risk factor because long-run abnormal return tests are inherently difficult. It also relies on Moody's ratings only, rather than both Moody's and Standard and Poor's. However, the authors argue that these limitations are unlikely to overturn the main result, given the size of the sample and the consistency of the downgrade evidence across tests.

12.6 Takeaway

The takeaway is simple: bond downgrades are bad news that equity investors appear to process too slowly. The strongest evidence comes from small, noninvestment-grade parent firms, where the first-year negative abnormal return is very large and where later earnings announcements reveal additional market surprise.